What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419771272 is the best physical fit.

Forward replay tested 5 candidates; the runner-up fits at 0.19. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.32	**5** / 6	**92%**	**5**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **5 of 6** forcing variants
- Inside the origin area **24 h** before the image
- Ahead of the runner-up by a mean loss gap of **0.0902**
- The nearest vessel, **419400107**, 0.0047° away, is not supported

Candidates, ranked by physical fit

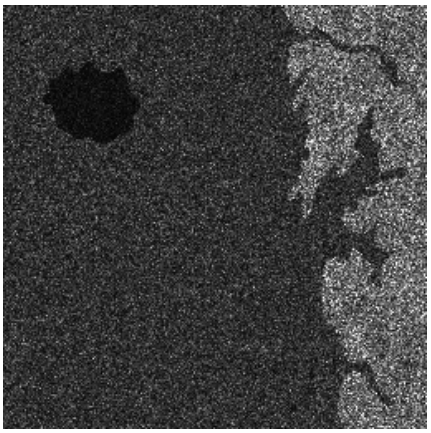
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419771272		0.325	5 of 6	9 Sep 2026, 20:30 UTC
2	419400107		0.190	1 of 6	10 Sep 2026, 04:00 UTC
3	419551858		0.035	0 of 6	9 Sep 2026, 14:30 UTC
4	419530006		0.000	0 of 6	9 Sep 2026, 22:30 UTC
5	419315497		0.000	0 of 6	9 Sep 2026, 22:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419400107**, ranked 2 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

92%
LEAD RETAINED

80 tested changes to the model's assumptions; **6** changed the leader.

The smallest change that flips it: **diffusivity multiplier 1.8** (magnitude 1.0), under which **419400107** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	483 km^2	0%	42%	0%
+12 h 10 Sep 2026, 16:30 UTC	537 km^2	0%	40%	0%
+24 h 11 Sep 2026, 04:30 UTC	578 km^2	0%	30%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

5 Candidate track data present and inside the backward envelope at some queried time

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**80%** 64/80	**80%** 70/88
False alerts	**25%** 32/128	**22%** 28/130
Mistaken promotions	**38%** 15/40	**57%** 23/40

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:10 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`13dd40672ab6431e8ffafce8434f229fb8aaecd4d7d304f649e127c3c0cde410`
coverage_result.json	`6ceb13a8c2e6da7d1295215e2d3e6c5bd1f68827a7f96dfbcb3cfadab8128dfe`
forcing.npz	`767ec8f532e8cc46ccabe89024236f1316cf88ae76649c1d650bc3106b859dbb`
forecast_result.json	`efb953534425a760c4310c6d92d41492521a49c62b8601eeb426401dcb88428d`
fragility_result.json	`42a592bcdf7fe525c3416115a0347308b07dd8de1f79ac22939b5de8d49c90d6`
historical_mapping.json	`748c15129fabdb2cdf8727c8bf0b91953bcba1ef24de5f9602c886399f783e44`
manifest.json	`1ff45cd57416e0cd75b8d7ba0fe3a69e9e935c299edf2bd31ac10f75c8d663c3`
mask_groundtruth.png	`f68e46a604d4c3c29220c1f05dceb16ee54227d5a6380b969c64b2a1d839b77d`
naive_baseline_result.json	`ec0c618119c3a020147a3eb13a6ad152e8c7e82f75db0c68dd1df0cf6f4c5ba2`
next_observation_result.json	`ae115bc6f214ab9b1aee26593d0b354bc285dc5dd1795a93b4cd2240cdafe60e`
scene.png	`2e9edc714c5336e2e5c71b9aea590131f3b7c688ff4fddd70bedc00ff1b33cce`

Uncertainty

— Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

— Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

— Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

— Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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